



Exercise for Unit 5

Name: _____

Date: _____

Name: _____

Year and Section: _____

Note: Save your Python source codes into a single .ipynb file OR multiple python files, representing each number, with the proper naming convention (see Readme on the repository), upload it to your assigned folder in the GitHub organization.

Training your own Skipgram. Train a word embedding model using the skip-gram architecture with negative sampling, then evaluate whether the learned vectors capture useful semantic relationships. Download the source code [here](#) and perform the following steps:

1. (10 points) Use a Wikipedia articles as your dataset:
 - a. Select a Wikipedia article of your choice as the corpora for this exercise.
 - b. The article should be reasonably long (at least a few thousand words) for good results.
 - c. Determine which part of the code (codeline) this part is:

Answer:

2. (10 points) Preprocess the text coming from the selected corpus.



- a. Determine which part of the code (codeline) this part is:

Answer:

3. (10 points) Train a Skip-gram with Negative Sampling model.

- Execute the code to train a skipgram.
- Take note of the following properties with Word2Vec, vector size and window.
- Determine which part of the code (codeline) this part is:

Answer:

4. (10 points) Evaluate the embeddings using a small test set.

- Evaluate the embedding output using a small test set.
- Change the list of words to be used for evaluation.
- Determine which part of the code (codeline) this part is:

Answer:

5. (10 points) Report nearest neighbors, similarity scores, and test-set performance.



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- a. Determine which part of the code (codeline) this part is:

Answer:



Answer the following questions:

1. (10 points) What are the critical parts of the script related on training the Word2Vec?

Answer:

2. (10 points) What Word2Vec methods were used to evaluate the similarity of words?

Answer:

Update the code to:

1. (15 points) Retrain the Word2Vec model with the following configuration:

- a. Window size of 10

Compare the evaluation values to the previous model. What can you derive from the change?

OLD:

NEW:



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2. (15 points) Visualize the vectors using PCA. Generate at least 20 known words and feed it on the visualization. Paste the graph below.

PCA: